

SENSOR-BASED IoT SIMULATION

CodeAlpha Internship – Task 2

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Domain: Internet of Things (IoT)

Platform: Tinkercad Circuits / Arduino

1. Project Title

Automatic Light Control Using an LDR Sensor and Arduino

2. Objective

The objective is to simulate a basic IoT sensing and control system in which an LDR detects surrounding light intensity and an Arduino Uno controls an LED according to the sensor reading. The project demonstrates sensing, processing and actuation without physical hardware.

3. Components Required

Component	Qty.	Purpose
Arduino Uno	1	Microcontroller
LDR / Photoresistor	1	Light sensing
10 k Ω resistor	1	Voltage divider
LED	1	Output indicator
220 Ω resistor	1	LED current limiting
Breadboard	1	Circuit assembly
Jumper wires	As required	Connections

4. Circuit Connections

Connect one LDR terminal to 5V and the other to A0. Connect a 10 k Ω resistor between A0 and GND, forming a voltage divider. Connect the LED anode to Arduino digital pin 9 through a 220 Ω resistor and the cathode to GND.

5. Arduino Program

```
const int LDR_PIN = A0;
const int LED_PIN = 9;
const int THRESHOLD = 500;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int lightValue = analogRead(LDR_PIN);
  Serial.println(lightValue);
  if (lightValue < THRESHOLD) digitalWrite(LED_PIN, HIGH);
  else digitalWrite(LED_PIN, LOW);
  delay(500);
}
```

6. Working Principle

The LDR changes its resistance according to the light falling on it. Arduino reads the resulting voltage at A0 and compares the reading with a threshold. When the detected light level is below the threshold, the LED turns ON;

when sufficient light is detected, the LED turns OFF.

7. Expected Output

Condition	Reading	LED
Low light / darkness	Below threshold	ON
Bright light	At/above threshold	OFF

8. Simulation Procedure

1. Open Tinkercad Circuits and create a new circuit. 2. Add Arduino Uno, breadboard, LDR, resistors and LED. 3. Make the connections above. 4. Enter the Arduino program. 5. Start simulation. 6. Change the LDR light level. 7. Observe the LED and Serial Monitor. 8. Capture screenshots of the circuit and different operating conditions.

9. Screenshots to Include

Screenshot 1: Complete circuit.

Screenshot 2: Low-light condition with LED ON.

Screenshot 3: Bright-light condition with LED OFF.

Screenshot 4: Code/Serial Monitor showing sensor values.

10. Result and Conclusion

The simulation demonstrates a basic automated sensing and control system. The LDR provides input, Arduino processes the sensor value, and the LED provides the output. The concept can be extended to smart lighting and other IoT applications by adding communication and cloud monitoring.